certain limit. The relationship between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) turned out to be the rectangular hyperbola. On a logarithmic scale, the relationship is a straight line described by the equation.

$$\log S = \log C + Z \log A$$

where

S = Species richness
A = Area
Z = Slope of the line (regression coefficient)
C = Y-intercept

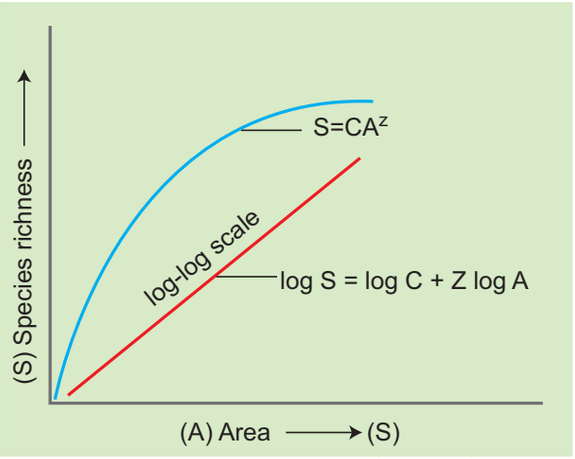



**Fig. 12.2** Species - Area relationship on log scale

Regression coefficient Z generally has a value of 0.1-0.2 regardless of taxonomic group or region. However, in case of the species – area relationship in very large areas like entire continents, the slope of the line appears to be much steeper (Z-value in the range of 0.6-1.2). For example, in case of the fruit eating (frugivorous) birds and mammals in the tropical forests of different continents, the slope is found to be a steeper line of 1.15 **(Fig. 12.2).**

## 12.2 Importance of biodiversity – Global and India

Biodiversity is the variety of life on earth. That is, it is the number of different species of flora and fauna including microorganisms. These organisms can inhabit different ecosystems with varying conditions like the Rainforests, Coral reefs, Grasslands, Deserts, Tundra and the Polar ice caps. This variety (Biodiversity) is essential for the wellbeing of our planet and sustenance of life as a whole.

Ecologist Paul Ehrlich proposed the 'Rivet Popper Hypothesis' for better understanding the loss of each species in the ecosystem. He compared each species of an ecosystem with rivets in the body of an aeroplane. Thousands of rivets (species) join all the parts of an aeroplane (ecosystem). If every passenger travelling in the aeroplane starts taking rivets home(loss of species), initially it may not affect flight safety (proper functioning of the ecosystem). However, the plane becomes dangerously weak over a period of time, when more and more rivets are removed. Moreover, which rivet is removed is also important. When the key rivets (removal of key stone species) on the wings of the aeroplane are removed, undoubtedly it poses a serious threat to the flight safety. Thus we understand the role of every species for the harmonial function of an ecosystem.

The importance of biodiversity can be viewed and measured as

a) Ecosystem services

b) Biological resources

c) Social benefits of biodiversity

The organization and functioning of ecosystems world over is effected and dependent on biodiversity and its richness. The major functional attributes are:

- continuity of nutrient cycles or biogeochemical cycles ($N_2$, C, $H_2O$, P, S cycles)
- soil formation, conditioning or maintenance of soil health (fertility) by soil microbial diversity along with the different trophic members



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